NEXUS PDB V2.0

Power copper simulation: current, temperature and pulse capability

Board revision: **PDB_HW_FILES commit 6ba6dcd** (4 Nov 2026) · 2 layers, 4 oz (140 µm) copper, 1.6 mm FR4 · 12S battery, 50.4 V max

Summary

- **100 A continuous is achievable, but not in still air in a warm enclosure.** With natural convection only, the board averages **+32 K** above ambient and the hottest spots reach **+65 K**: about **90 °C at 25 °C ambient** and **105 °C at 40 °C**. A small fan (~1 m/s across the board) brings the peak down to **+36 K**.
- Total heat at 100 A is **20.8 W**: 10.6 W in the copper, 6.4 W in the four MOSFETs and about 3.8 W in connector contacts.
- Copper resistance from the battery connector, through the MOSFET pads, to the outputs and back is **1.15 mΩ**. That is a 115 mV drop at 100 A and 344 mV at 300 A, not counting the MOSFETs.
- **300 A pulses of a few milliseconds are not a problem.** A 10 ms pulse adds at most **+11 K** at the worst spot in the copper. Pulses up to about 20 ms stay under +22 K. Avoid 300 A for longer than about 50 ms (+53 K at the worst spot).
- The hottest areas are the **MOSFET tabs (U14 most)**, the **J13 battery input**, and a **5 mm-wide neck in the bottom-layer BATT+ copper** between J13 and the MOSFETs, near (265, 239) mm. In a 40 °C enclosure with still air, the board under the **85 °C-rated bulk capacitors C1-C4 reaches about 80 °C**, and under **C5 (105 °C) about 99 °C**, which is too close to their ratings (Table 4).

Case	Peak rise (K)	Peak at 25 °C amb.	Peak at 40 °C amb.	Board average rise (K)
100 A, still air	65	90 °C	105 °C	32
100 A, light airflow	36	61 °C	76 °C	12
+ 300 A for 10 ms on top	+11 at worst spot			

Table 1. Key results.

1. What was simulated

The board file was read directly (zone fills, tracks, pads and vias). Each power net was turned into a fine grid of copper cells (0.25 mm) on both layers, joined by every via and through-hole pad, and solved for current flow. The heat produced in every cell was then fed into a second model of the whole board (both copper layers, the FR4 between them, vias, and heat loss to air from both faces) to get the steady temperature. The 300 A pulse was handled separately, because in a few milliseconds heat cannot spread far: the copper heats up in place.

Item	Value used	Note
Copper	4 oz (140 µm) both layers, resistivity at ~70 °C	conservative for a warm board
Current	100 A continuous; 300 A pulse	battery in at J13, out through the 12 XT60s
Load split	8.3 A per output (all 12 equal); 25 A per MOSFET	a real robot loads outputs unequally
MOSFET loss	2.55 mΩ each (1.7 mΩ max × 1.5 for hot)	1.59 W each at 25 A
Contact resistance	QS8 0.15 mΩ per contact, XT60 0.5 mΩ per contact	assumed; measure on real parts
Cooling (still air)	top 17 W/m²K, bottom 12 W/m²K	natural convection + radiation, board horizontal
Cooling (airflow)	37 W/m²K both faces	about 1 m/s across the board
Not included	heat from the 5 V converter, logic and precharge parts; enclosure walls; cables carrying heat away	cables usually help

Table 2. Assumptions.

2. Board copper

Figure 1 shows which copper belongs to which power net. Blue is the battery side before the switch (BATT+), orange is the switched output bus (BUS+), and green is the return (GND).

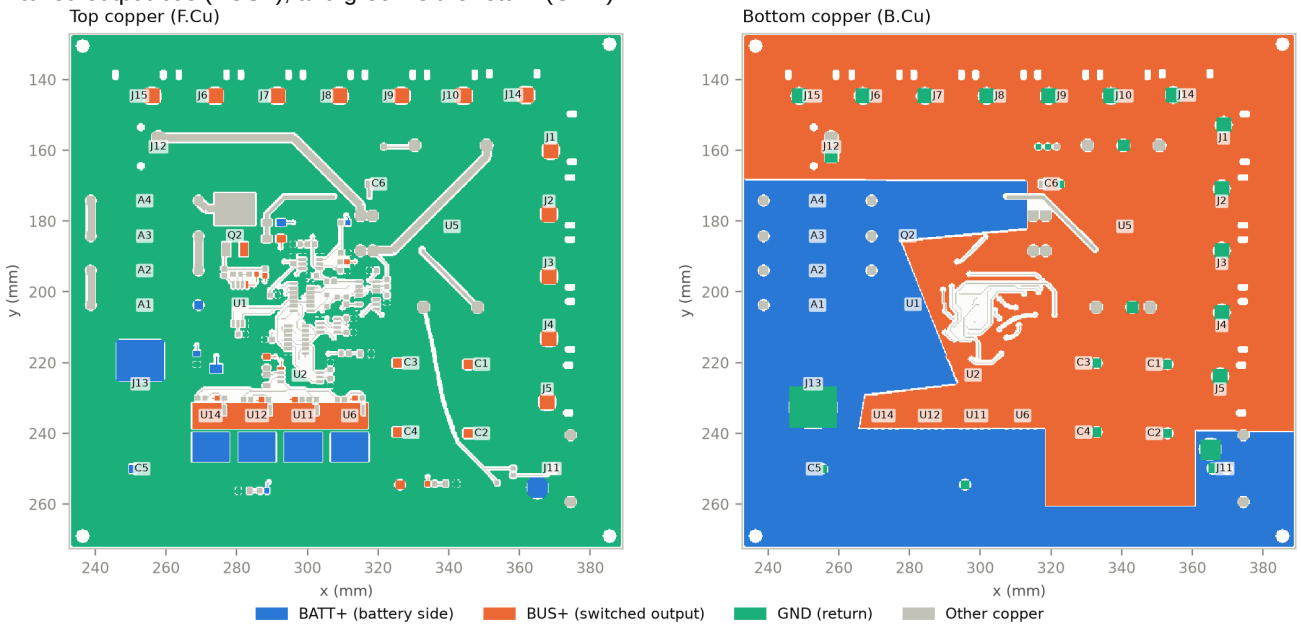


Figure 1. Power nets on the top and bottom copper, with the main parts labelled.

3. Current density at 100 A

Current density shows how hard each part of the copper is working. For 4 oz copper, below about **25 A/mm²** is comfortable for continuous current (orange contour line). Above that is acceptable in small areas when the surrounding copper spreads the heat. Dark areas are where the current crowds.

Net	Path	Resistance	Drop 100 A	Drop 300 A	Heat 100 A	Heat 300 A
BATT+	J13 to MOSFET tabs	379 µΩ	38 mV	114 mV	3.4 W	30 W
BUS+	MOSFETs to 12 outputs	367 µΩ	37 mV	110 mV	3.5 W	31 W
GND	12 outputs to J13	401 µΩ	40 mV	120 mV	3.7 W	34 W
Total		1148 µΩ	115 mV	344 mV	10.6 W	95 W

Table 3. Copper resistance, voltage drop and heat per net. At 300 A the heat is 9 times higher, but only for the length of the pulse.

BATT+ : J13 to the four MOSFET tabs

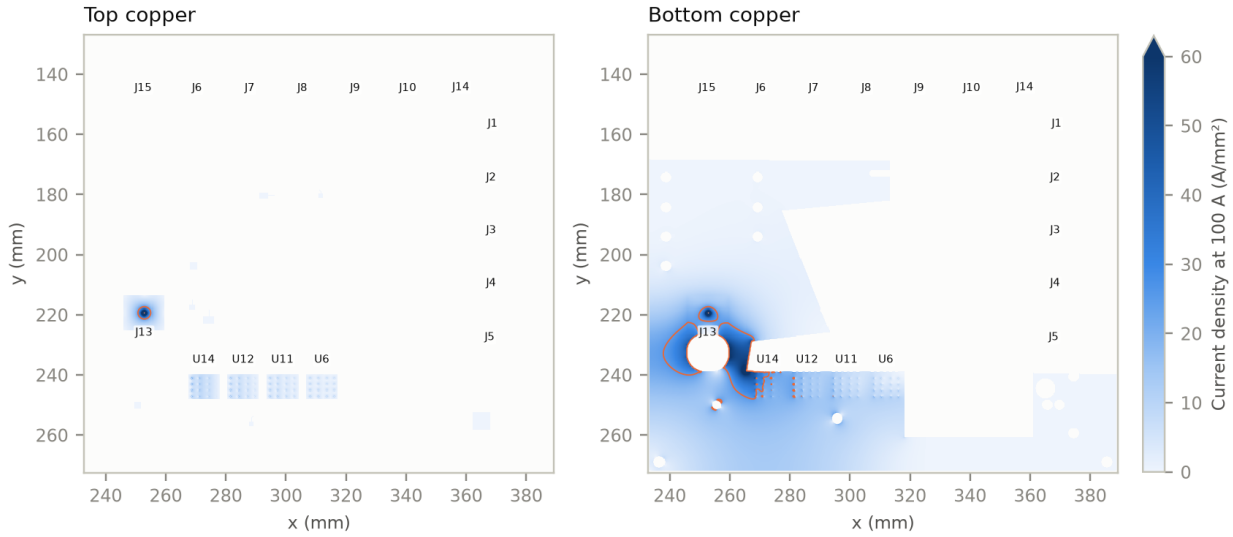


Figure 2. BATT+ current density. The current has to pass around the J13 GND pin to reach the MOSFETs, through a neck about 5 mm wide.

BUS+ : MOSFET sources to the 12 XT60 outputs

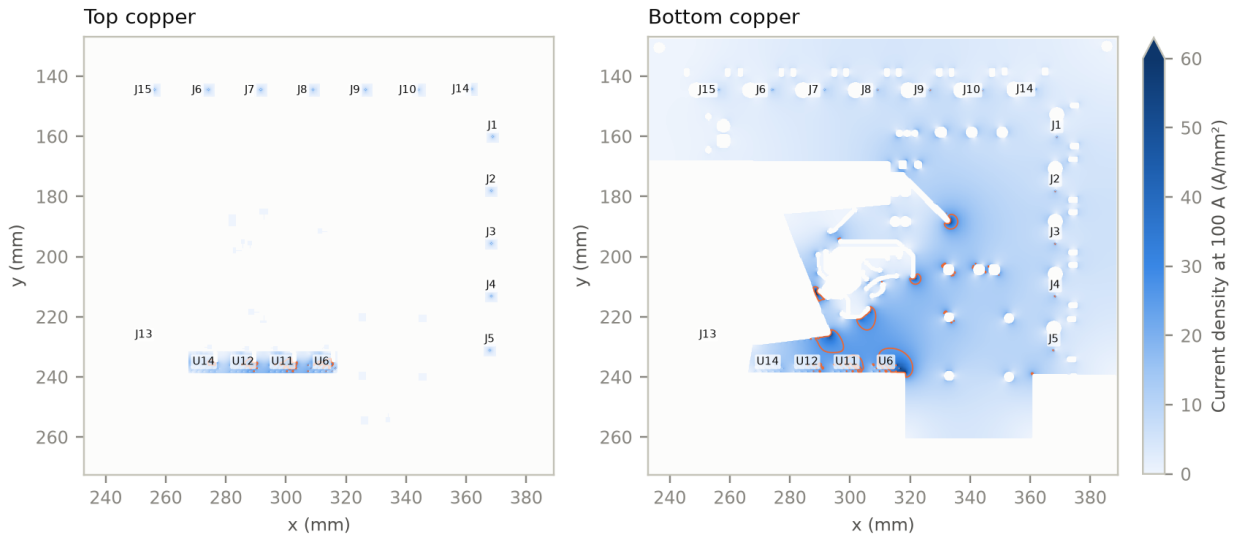


Figure 3. BUS+ current density. Spread well over both layers.

GND : 12 XT60 returns to J13

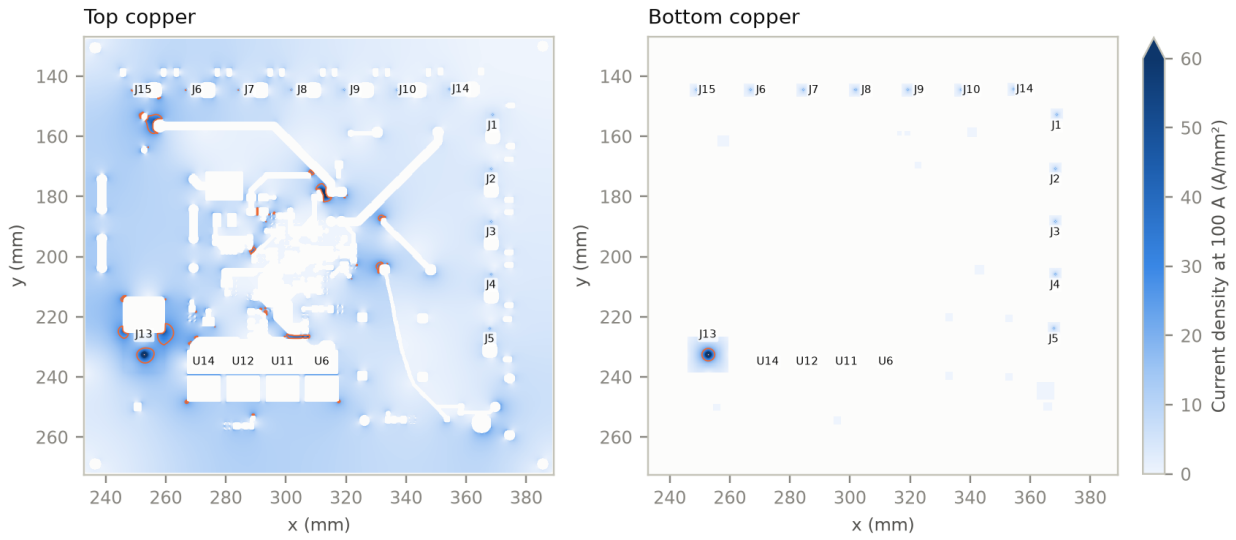


Figure 4. GND current density. Crowding is only right next to J13 and a few output pads.

4. Temperature at 100 A continuous

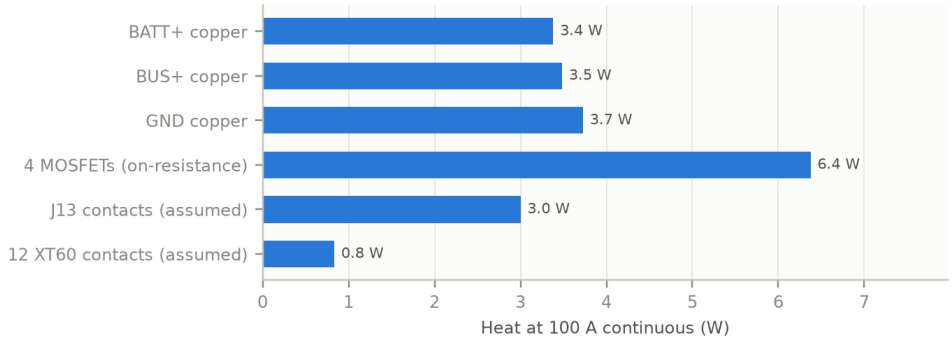


Figure 5. Where the heat comes from at 100 A.

Figure 6 shows the steady temperature in still air at 25 °C ambient. Circles mark the hottest spots. For a 40 °C enclosure, add 15 °C to every value.

100 A continuous, still air (natural convection)

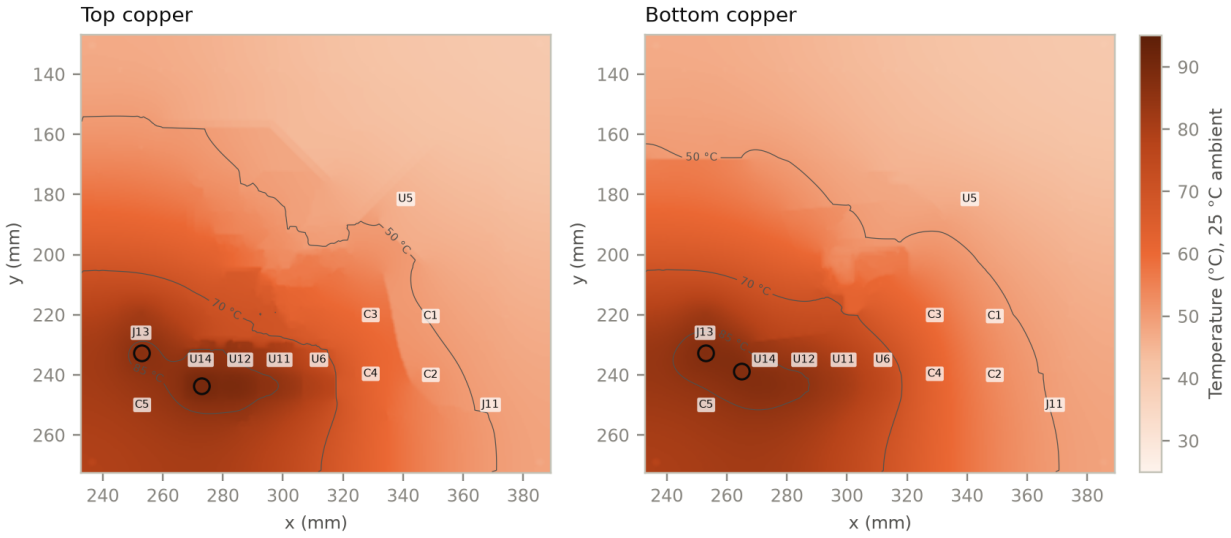


Figure 6. Steady temperature at 100 A continuous, still air, 25 °C ambient. Contour lines at 50, 70 and 85 °C.

Part	What it is	Still air, 25 °C amb.	Still air, 40 °C amb.	Airflow, 40 °C amb.	Rating
U6, U11, U12, U14	Main MOSFETs (tab)	90 °C	105 °C	76 °C	175 °C junction
J13	QS8 battery input	88 °C	103 °C	74 °C	connector housing
C1-C4	Bulk 1000 µF 100 V (Nichicon VR)	65 °C	80 °C	58 °C	85 °C
C5	100 µF 100 V (Panasonic FC)	84 °C	99 °C	69 °C	105 °C
C6	100 µF 100 V (Panasonic FC)	45 °C	60 °C	46 °C	105 °C
U5	URB4805 5 V converter	54 °C	69 °C	51 °C	see datasheet derating
U1, U2	TPSI3050 drivers	76 °C	91 °C	65 °C	125 °C
R1_A1-A4	Precharge resistors	70 °C	85 °C	60 °C	resistor rating
J1-J15	XT60 outputs (hottest)	49 °C	64 °C	48 °C	connector housing

Table 4. Board temperature under each part (hottest point under its pads). The part itself runs a few degrees hotter than the board under it. Ratings are from the parts in the BOM; check each datasheet.

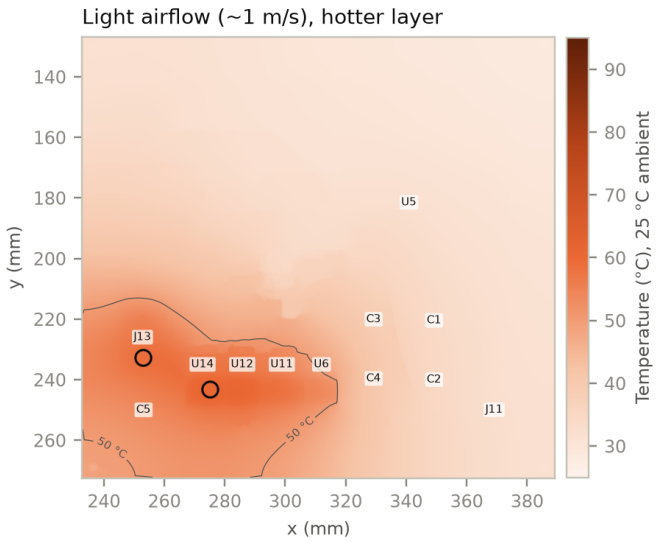


Figure 7. The same 100 A with light airflow.

Same 20.8 W of heat, but with a small fan blowing across both sides of the board.

Peak rise falls from 65 K to 36 K and the board average from 32 K to 12 K.

5. 300 A for a few milliseconds

In a short pulse the heat has no time to spread (in 10 ms it moves only about 1 mm in copper), so each piece of copper heats up by itself. The extra rise grows in proportion to pulse length and to the current squared. The result below is added on top of the 100 A steady temperature. Areas right at the connector and MOSFET pads are greyed out, because there the model forces current evenly into the whole pad, which is not how the real pin carries it.

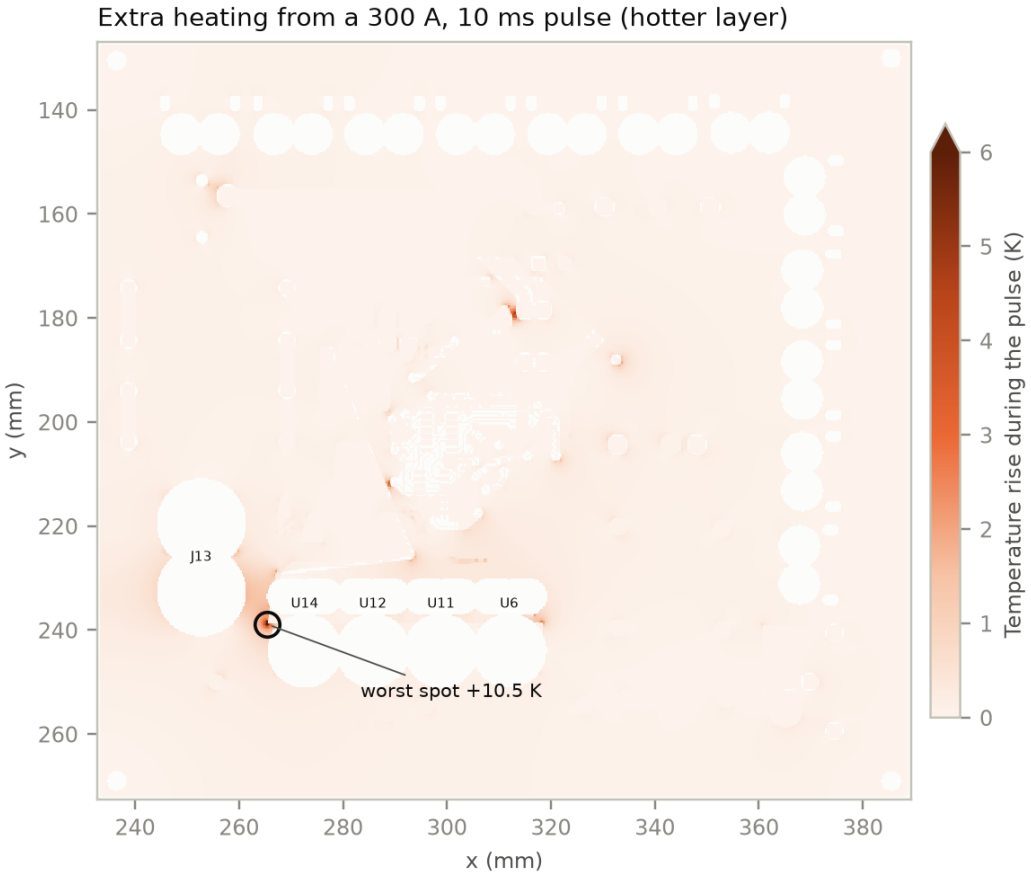


Figure 8. Extra temperature rise from a 300 A pulse lasting 10 ms.

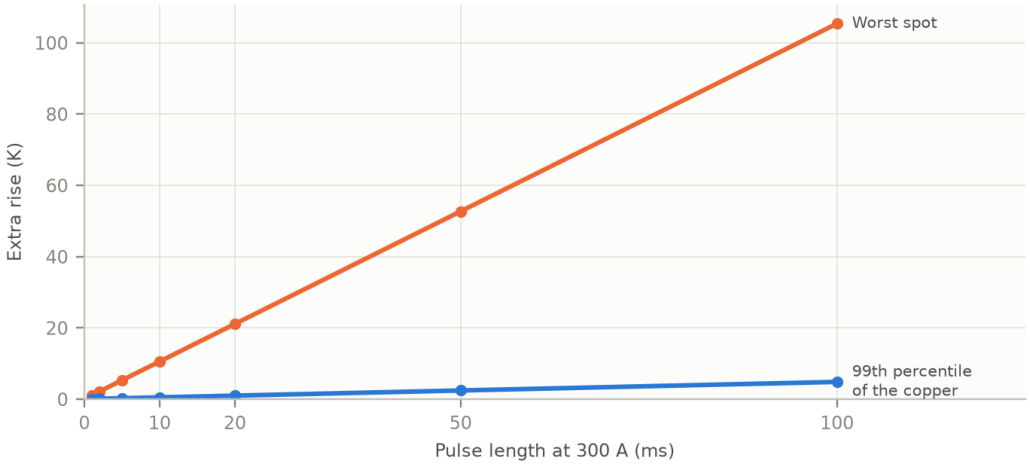


Figure 9. Extra rise against pulse length at 300 A.

Pulse at 300 A	1 ms	5 ms	10 ms	20 ms	50 ms	100 ms
Worst spot (K)	+1.1	+5.3	+10.5	+21.1	+52.7	+105.3
99th percentile (K)	+0.0	+0.2	+0.5	+1.0	+2.4	+4.8

Table 5. Extra rise in the copper during a 300 A pulse (no heat spreading, so these are upper limits).

MOSFETs during the pulse: each carries 75 A and dissipates about 14 W. Even using the steady junction-to-case resistance (about 0.4 K/W), which overstates a millisecond pulse, the junction rises less than 6 K. The voltage drop across the board copper at 300 A is about 344 mV, plus about 191 mV across the MOSFETs.

6. Findings and recommendations

- **Plan cooling for 100 A continuous.** In still air inside a warm robot body, the MOSFET area and J13 reach about 105 °C. That is within the MOSFET and FR4 limits, but it is hot for neighbouring parts and connectors. A small fan or a metal heat-spreader under the MOSFETs cuts the peak rise roughly in half.
- **Upgrade or move the electrolytic capacitors.** C1-C4 are 85 °C parts and reach about 80 °C (still air, 40 °C ambient); every 10 °C hotter roughly halves the life of an electrolytic capacitor. Use 105 °C parts of the same size, or add airflow. C5 sits next to J13 and reaches about 99 °C against its 105 °C rating; move it away from J13 if possible.
- **Optional: rotate J13 by 180°** so the BATT+ pin faces the MOSFETs. That removes the 5 mm neck in Figure 2 and its hot spot.
- **Measure J13 contact resistance.** The 3 W assumed here at the QS8 is a guess. A poor or worn anti-spark connector can make J13 the hottest point on the board.
- **300 A pulses are fine up to tens of milliseconds.** The limit for longer peaks is the off-board fuse and the battery, not the PCB copper.
- **Unequal loads.** If two or three outputs carry most of the current (for example, the knees during a squat), the copper next to those XT60s runs hotter than shown here. Re-run with real per-actuator currents once they are known.

7. Limits of this model

This is an engineering estimate, not a measurement. It treats the board as flat and horizontal in free air, does not include heat carried away by the cables (which usually helps), and uses assumed contact resistances and cooling coefficients. Results should be within about 20-30 % for temperature rise. Confirm on the first board with a thermocouple or thermal camera at 50 A, then scale up: temperature rise grows roughly with the square of the current.